



Chapter 3: Introduction to SQL

Database System Concepts, 7th Ed.

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Outline

- Overview of The SQL Query Language
- SQL Data Definition
- Basic Query Structure of SQL Queries
- Additional Basic Operations
- Set Operations
- Null Values
- Aggregate Functions
- Nested Subqueries
- Modification of the Database



History

- IBM Sequel language developed as part of System R project at the IBM San Jose Research Laboratory
- Renamed **Structured Query Language (SQL)**
- ANSI and ISO standard SQL:
 - SQL-86
 - SQL-89
 - SQL-92
 - SQL:1999 (language name became Y2K compliant!)
 - SQL:2003
- Commercial systems offer most, if not all, SQL-92 features, plus varying feature sets from later standards and special proprietary features.
 - Not all examples here may work on your particular system.



SQL Parts

DML (Data Manipulation Language): Provides commands to query, insert, update, and delete data within the database.

Integrity: Enables the specification of rules (constraints) to ensure the accuracy and consistency of data, such as enforcing valid data ranges or unique values.

View Definition: Includes commands to create views, which are virtual tables derived from existing tables for simplified and restricted data access.

Transaction Control: Facilitates managing transactions using commands to begin, commit (save changes), or rollback (revert changes) operations.

Embedded and Dynamic SQL: Defines how SQL commands can be integrated into general-purpose programming languages for seamless database interaction.

Authorization: Provides commands to define and manage user permissions, specifying access rights to data and views to ensure security.



Data Definition Language

SQL Data Definition Language (DDL) is used to define the structure of a database. It helps you specify details about the tables (or relations) in the database, such as:

The Schema: This is the blueprint or design of the table, describing its structure (like the columns and their data types).

Attribute Types: This specifies what kind of data each column can store (for example, text, numbers, dates).

Integrity Constraints: These are rules that ensure the data is correct and valid, like preventing duplicate entries or ensuring certain fields are not empty.

Indexes: Indexes are special structures that speed up data search and retrieval in the table.

Security and Permissions: This defines who can access or modify the data in the table, and what actions they can perform.

Storage Details: This refers to how the data is actually stored on the disk, like organizing it for efficiency and performance.



Domain Types in SQL

- **char(*n*).** Fixed length character string, with user-specified length *n*.
- **varchar(*n*).** Variable length character strings, with user-specified maximum length *n*.
- **int.** Integer (a finite subset of the integers that is machine-dependent).
- **smallint.** Small integer (a machine-dependent subset of the integer domain type).
- **numeric(*p*,*d*).** Fixed point number, with user-specified precision of *p* digits, with *d* digits to the right of decimal point. (ex., **numeric(3,1)**, allows 44.5 to be stored exactly, but not 444.5 or 0.32)
- **real, double precision.** Floating point and double-precision floating point numbers, with machine-dependent precision.
- **float(*n*).** Floating point number, with user-specified precision of at least *n* digits.
- More are covered in Chapter 4.

Domain refers to the set of valid values that a column (attribute) can have. It defines constraints on the type of data that can be stored in a specific field



Create Table Construct

- An SQL relation is defined using the **create table** command:

create table *r*

$(A_1 D_1, A_2 D_2, \dots, A_n D_n,$
 (integrity-constraint₁),
 ...,
 (integrity-constraint_k))

- *r* is the name of the relation
 - each A_i is an attribute name in the schema of relation *r*
 - D_i is the data type of values in the domain of attribute A_i
- Example:

```
create table instructor (  
    ID           char(5),  
    name        varchar(20),  
    dept_name   varchar(20),  
    salary     numeric(8,2))
```



Integrity Constraints in Create Table

- Types of integrity constraints
 - primary key** ($A_1, \dots, A_n$)
 - foreign key** ($A_m, \dots, A_n$) **references** r
 - not null**
- SQL prevents any update to the database that violates an integrity constraint.
- Example:

```
create table instructor (  
    ID          char(5),  
    name        varchar(20) not null,  
    dept_name varchar(20),  
    salary     numeric(8,2),  
    primary key (ID),  
    foreign key (dept_name) references department);
```

In a database, constraints are rules applied to the data in a table to ensure its integrity, accuracy, and reliability.



And a Few More Relation Definitions

- **create table** *student* (
 ID **varchar**(5),
 name **varchar**(20) not null,
 dept_name **varchar**(20),
 tot_cred **numeric**(3,0),
 primary key (*ID*),
 foreign key (*dept_name*) **references** *department*);

- **create table** *takes* (
 ID **varchar**(5),
 course_id **varchar**(8),
 sec_id **varchar**(8),
 semester **varchar**(6),
 year **numeric**(4,0),
 grade **varchar**(2),
 primary key (*ID*, *course_id*, *sec_id*, *semester*, *year*) ,
 foreign key (*ID*) **references** *student*,
 foreign key (*course_id*, *sec_id*, *semester*, *year*) **references** *section*);



And more still

- **create table** *course* (
 course_id **varchar**(8),
 title **varchar**(50),
 dept_name **varchar**(20),
 credits **numeric**(2,0),
 primary key (*course_id*),
 foreign key (*dept_name*) **references** *department*);



Updates to tables

- **Insert**
 - **insert into** *instructor* **values** ('10211', 'Smith', 'Biology', 66000);
- **Delete**
 - Remove all tuples from the *student* relation
 - **delete from** *student*
- **Drop Table**
 - **drop table** *r*
- **Alter** The ALTER keyword in a database is used to modify the structure of an existing table without losing data.
 - **alter table** *r add A D*
 - where *A* is the name of the attribute to be added to relation *r* and *D* is the domain of *A*.
 - All exiting tuples in the relation are assigned *null* as the value for the new attribute.
 - **alter table** *r drop A*
 - where *A* is the name of an attribute of relation *r*
 - Dropping of attributes not supported by many databases.



Basic Query Structure

- A typical SQL query has the form:

select $A_1, A_2, \dots, A_n$
from $r_1, r_2, \dots, r_m$
where P

- A_i represents an attribute
 - R_i represents a relation
 - P is a predicate.
- The result of an SQL query is a relation.



The select Clause

- The SELECT clause in SQL is used to specify which columns (attributes) you want to include in the result of a query.

This is similar to the projection operation in relational algebra, which selects certain columns from a table.

- Example: find the names of all instructors:

```
select name  
from instructor
```

- NOTE: SQL names are case insensitive (i.e., you may use upper- or lower-case letters.)
 - E.g., *Name* $\equiv$ *NAME* $\equiv$ *name*
 - Some people use upper case wherever we use bold font.



The select Clause (Cont.)

- SQL allows duplicates in relations as well as in query results.
- To force the elimination of duplicates, insert the keyword **distinct** after select.
- Find the department names of all instructors, and remove duplicates

```
select distinct dept_name  
from instructor
```

- The keyword **all** specifies that duplicates should not be removed.

```
select all dept_name  
from instructor
```



The select Clause (Cont.)

- An asterisk in the select clause denotes “all attributes”

```
select *  
from instructor
```

- An attribute can be a literal with no **from** clause

```
select '437'
```

- Results is a table with one column and a single row with value “437”
- Can give the column a name using:

```
select '437' as FOO
```

- An attribute can be a literal with **from** clause

```
select 'A'  
from instructor
```

- Result is a table with one column and N rows (number of tuples in the *instructors* table), each row with value “A”



The select Clause (Cont.)

- The **select** clause can contain arithmetic expressions involving the operation, +, −, *, and /, and operating on constants or attributes of tuples.

- The query:

```
select ID, name, salary/12  
from instructor
```

would return a relation that is the same as the *instructor* relation, except that the value of the attribute *salary* is divided by 12.

- Can rename “salary/12” using the **as** clause:

```
select ID, name, salary/12 as monthly_salary
```



The where Clause

- The **where** clause specifies conditions that the result must satisfy
 - Corresponds to the selection predicate of the relational algebra.
- To find all instructors in Comp. Sci. dept

```
select name  
from instructor  
where dept_name = 'Comp. Sci.'
```

- SQL allows the use of the logical connectives **and**, **or**, and **not**
- The operands of the logical connectives can be expressions involving the comparison operators <, <=, >, >=, =, and <>.
- Comparisons can be applied to results of arithmetic expressions
- To find all instructors in Comp. Sci. dept with salary > 80000

```
select name  
from instructor  
where dept_name = 'Comp. Sci.' and salary > 80000
```



The from Clause

- The **from** clause lists the relations involved in the query
 - Corresponds to the Cartesian product operation of the relational algebra.

- Find the Cartesian product *instructor X teaches*

select *
from *instructor, teaches*

- generates every possible instructor – teaches pair, with all attributes from both relations.
 - For common attributes (e.g., *ID*), the attributes in the resulting table are renamed using the relation name (e.g., *instructor.ID*)
- Cartesian product not very useful directly, but useful combined with where-clause condition (selection operation in relational algebra).



Examples

- Find the names of all instructors who have taught some course and the course_id
 - **select** *name, course_id*
from *instructor , teaches*
where *instructor.ID = teaches.ID*

- Find the names of all instructors in the Art department who have taught some course and the course_id
 - **select** *name, course_id*
from *instructor , teaches*
where *instructor.ID = teaches.ID and instructor. dept_name = 'Art'*



The Rename Operation

- The SQL allows renaming relations and attributes using the **as** clause:

old-name as new-name

- Find the names of all instructors who have a higher salary than some instructor in 'Comp. Sci'.
 - **select distinct** *T.name*
from *instructor as T, instructor as S*
where *T.salary > S.salary and S.dept_name = 'Comp. Sci.'*
- Keyword **as** is optional and may be omitted
instructor as T $\equiv$ *instructor T*



Self Join Example

- Relation *emp-super*

<i>person</i>	<i>supervisor</i>
Bob	Alice
Mary	Susan
Alice	David
David	Mary

- Find the supervisor of “Bob”
- Find the supervisor of the supervisor of “Bob”
- Can you find ALL the supervisors (direct and indirect) of “Bob”?

1. Find the direct supervisor of "Bob":

sql

```
SELECT supervisor FROM emp_super WHERE person = 'Bob';
```

2. Find the supervisor of the supervisor of "Bob":

sql

```
SELECT supervisor
FROM emp_super
WHERE person = (SELECT supervisor FROM emp_super WHERE person = 'Bob');
```

3. Find ALL supervisors (direct and indirect) of "Bob":

sql

```
WITH SupervisorHierarchy AS (
    SELECT person, supervisor FROM emp_super WHERE person = 'Bob'
    UNION ALL
    SELECT e.person, s.supervisor
    FROM emp_super e
    JOIN SupervisorHierarchy s ON e.supervisor = s.person
)
SELECT supervisor FROM SupervisorHierarchy;
```



String Operations

- SQL includes a string-matching operator for comparisons on character strings. The operator **like** uses patterns that are described using two special characters:
 - percent (%). The % character matches any substring.
 - underscore (_). The _ character matches any character.
- Find the names of all instructors whose name includes the substring “dar”.

```
select name  
from instructor  
where name like '%dar%'
```

- Match the string “100%”

```
like '100 \%' escape '\'
```

in that above we use backslash (\) as the escape character.



String Operations (Cont.)

- Patterns are case sensitive.
- Pattern matching examples:
 - 'Intro%' matches any string beginning with “Intro”.
 - '%Comp%' matches any string containing “Comp” as a substring.
 - '___' matches any string of exactly three characters.
 - '___%' matches any string of at least three characters.
- SQL supports a variety of string operations such as
 - concatenation (using “||”)
 - converting from upper to lower case (and vice versa)
 - finding string length, extracting substrings, etc.



Ordering the Display of Tuples

- List in **alphabetic order** the names of all instructors

```
select distinct name  
from   instructor  
order by name
```

- We may specify **desc** for descending order or **asc** for ascending order, for each attribute; ascending order is the default.
 - Example: **order by** *name desc*
- Can sort on multiple attributes
 - Example: **order by** *dept_name, name*



Where Clause Predicates

- SQL includes a **between** comparison operator
- Example: Find the names of all instructors with salary between \$90,000 and \$100,000 (that is, $\geq \$90,000$ and $\leq \$100,000$)
 - **select** *name*
from *instructor*
where *salary* **between** 90000 **and** 100000
- Tuple comparison
 - **select** *name, course_id*
from *instructor, teaches*
where (*instructor.ID, dept_name*) = (*teaches.ID*, 'Biology');



Set Operations

- Find courses that ran in Fall 2017 or in Spring 2018

```
(select course_id from section where sem = 'Fall' and year = 2017)  
union  
(select course_id from section where sem = 'Spring' and year = 2018)
```
- Find courses that ran in Fall 2017 and in Spring 2018

```
(select course_id from section where sem = 'Fall' and year = 2017)  
intersect  
(select course_id from section where sem = 'Spring' and year = 2018)
```
- Find courses that ran in Fall 2017 but not in Spring 2018

```
(select course_id from section where sem = 'Fall' and year = 2017)  
except  
(select course_id from section where sem = 'Spring' and year = 2018)
```



Set Operations (Cont.)

- Set operations **union, intersect, and except**
 - Each of the above operations automatically eliminates duplicates
- To **retain all duplicates** use the
 - **union all,**
 - **intersect all**
 - **except all.**



Null Values

- It is possible for tuples to have a null value, denoted by **null**, for some of their attributes
- **null signifies an unknown value or that a value does not exist.**
- The result of any arithmetic expression involving **null** is **null**
 - Example: $5 + \text{null}$ returns **null**
- The predicate **is null** can be used to check for null values.
 - Example: Find all instructors whose salary is null.
select name
from instructor
where salary is null
- The predicate **is not null** succeeds if the value on which it is applied is not null.



Null Values (Cont.)

- SQL treats as **unknown** the result of any comparison involving a null value (other than predicates **is null** and **is not null**).
 - Example: $5 < \text{null}$ or $\text{null} <> \text{null}$ or $\text{null} = \text{null}$
- The predicate in a **where** clause can involve Boolean operations (**and**, **or**, **not**); thus the definitions of the Boolean operations need to be extended to deal with the value **unknown**.
 - **and** : $(\text{true and unknown}) = \text{unknown}$,
 $(\text{false and unknown}) = \text{false}$,
 $(\text{unknown and unknown}) = \text{unknown}$
 - **or**: $(\text{unknown or true}) = \text{true}$,
 $(\text{unknown or false}) = \text{unknown}$
 $(\text{unknown or unknown}) = \text{unknown}$
- Result of **where** clause predicate is treated as *false* if it evaluates to *unknown*



Aggregate Functions

- Aggregate functions in SQL are used to perform calculations on a set of values and return a single value

avg: average value

min: minimum value

max: maximum value

sum: sum of values

count: number of values



Aggregate Functions Examples

- Find the average salary of instructors in the Computer Science department
 - **select avg (*salary*)**
from *instructor*
where *dept_name* = 'Comp. Sci.';
- Find the total number of instructors who teach a course in the Spring 2018 semester
 - **select count (distinct *ID*)**
from *teaches*
where *semester* = 'Spring' and *year* = 2018;
- Find the number of tuples in the *course* relation
 - **select count (*)**
from *course*;



Aggregate Functions – Group By

- Find the average salary of instructors in each department
 - select** *dept_name*, **avg** (*salary*) **as** *avg_salary*
from *instructor*
group by *dept_name*;

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
76766	Crick	Biology	72000
45565	Katz	Comp. Sci.	75000
10101	Srinivasan	Comp. Sci.	65000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000
12121	Wu	Finance	90000
76543	Singh	Finance	80000
32343	El Said	History	60000
58583	Califieri	History	62000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
22222	Einstein	Physics	95000

<i>dept_name</i>	<i>avg_salary</i>
Biology	72000
Comp. Sci.	77333
Elec. Eng.	80000
Finance	85000
History	61000
Music	40000
Physics	91000



Aggregation (Cont.)

- In SQL, the GROUP BY clause is used to arrange identical data into groups. It is often used in conjunction with aggregate functions like COUNT(), SUM(), AVG(), MAX(), and MIN() to perform calculations on each group of data.

Example

Suppose you have a table named Sales with the following columns: ProductID, Quantity, and SaleDate. You want to find the total quantity sold for each product.

```
SELECT ProductID, SUM(Quantity) AS TotalQuantity  
FROM Sales  
GROUP BY ProductID;
```



Aggregate Functions – Having Clause

- Find the names and average salaries of all departments whose average salary is greater than 42000

```
select dept_name, avg (salary) as avg_salary  
from instructor  
group by dept_name  
having avg (salary) > 42000;
```

- Note: predicates in the **having** clause are applied after the formation of groups whereas predicates in the **where** clause are applied before forming groups



Nested Subqueries

- SQL provides a mechanism for the nesting of subqueries. A **subquery** is a **select-from-where** expression that is nested within another query.
- The nesting can be done in the following SQL query

```
select  $A_1, A_2, \dots, A_n$   
from  $r_1, r_2, \dots, r_m$   
where  $P$ 
```

as follows:

- **From clause:** r_i can be replaced by any valid subquery
- **Where clause:** P can be replaced with an expression of the form:
$$B \text{ <operation> (subquery)}$$
$$B$$
 is an attribute and <operation> to be defined later.
- **Select clause:**
 A_i can be replaced by a subquery that generates a single value.



Set Membership



Set Membership

- Find courses offered in Fall 2017 and in Spring 2018

```
select distinct course_id
from section
where semester = 'Fall' and year= 2017 and
       course_id in (select course_id
                      from section
                      where semester = 'Spring' and year= 2018);
```

- Find courses offered in Fall 2017 but not in Spring 2018

```
select distinct course_id
from section
where semester = 'Fall' and year= 2017 and
       course_id not in (select course_id
                      from section
                      where semester = 'Spring' and year= 2018);
```



Set Membership (Cont.)

- Name all instructors whose name is neither “Mozart” nor Einstein”

```
select distinct name  
from instructor  
where name not in ('Mozart', 'Einstein')
```

- Find the total number of (distinct) students who have taken course sections taught by the instructor with *ID* 10101

```
select count (distinct ID)  
from takes  
where (course_id, sec_id, semester, year) in  
      (select course_id, sec_id, semester, year  
       from teaches  
       where teaches.ID= 10101);
```

- Note: Above query can be written in a much simpler manner.
The formulation above is simply to illustrate SQL features



Set Comparison



Set Comparison – “some” Clause

- Find names of instructors with salary greater than that of some (at least one) instructor in the Biology department.

```
select distinct T.name  
from instructor as T, instructor as S  
where T.salary > S.salary and S.dept name = 'Biology';
```

- Same query using > **some** clause

```
select name  
from instructor  
where salary > some (select salary  
                        from instructor  
                        where dept name = 'Biology');
```



Definition of “some” Clause

- $F \text{ <comp> some } r \Leftrightarrow \exists t \in r \text{ such that } (F \text{ <comp> } t)$

Where <comp> can be: <, ≤, >, =, ≠

$$(5 < \text{some } \begin{array}{|c|} \hline 0 \\ \hline 5 \\ \hline 6 \\ \hline \end{array}) = \text{true}$$

(read: 5 < some tuple in the relation)

$$(5 < \text{some } \begin{array}{|c|} \hline 0 \\ \hline 5 \\ \hline \end{array}) = \text{false}$$

$$(5 = \text{some } \begin{array}{|c|} \hline 0 \\ \hline 5 \\ \hline \end{array}) = \text{true}$$

$$(5 \neq \text{some } \begin{array}{|c|} \hline 0 \\ \hline 5 \\ \hline \end{array}) = \text{true (since } 0 \neq 5)$$

$(= \text{some}) \equiv \text{in}$

However, $(\neq \text{some}) \not\equiv \text{not in}$



Set Comparison – “all” Clause

- Find the names of all instructors whose salary is greater than the salary of all instructors in the Biology department.

```
select name
from instructor
where salary > all (select salary
                        from instructor
                        where dept name = 'Biology');
```



Definition of “all” Clause

- $F <\text{comp}> \text{all } r \Leftrightarrow \forall t \in r (F <\text{comp}> t)$

$$(5 < \text{all } \begin{array}{|c|} \hline 0 \\ \hline 5 \\ \hline 6 \\ \hline \end{array}) = \text{false}$$

$$(5 < \text{all } \begin{array}{|c|} \hline 6 \\ \hline 10 \\ \hline \end{array}) = \text{true}$$

$$(5 = \text{all } \begin{array}{|c|} \hline 4 \\ \hline 5 \\ \hline \end{array}) = \text{false}$$

$$(5 \neq \text{all } \begin{array}{|c|} \hline 4 \\ \hline 6 \\ \hline \end{array}) = \text{true (since } 5 \neq 4 \text{ and } 5 \neq 6)$$

$(\neq \text{all}) \equiv \text{not in}$

However, $(= \text{all}) \not\equiv \text{in}$



Test for Empty Relations

- The **exists** construct returns the value **true** if the argument subquery is nonempty.
- **exists** $r \Leftrightarrow r \neq \emptyset$
- **not exists** $r \Leftrightarrow r = \emptyset$



Use of “exists” Clause

- Yet another way of specifying the query “Find all courses taught in both the Fall 2017 semester and in the Spring 2018 semester”

```
select course_id
from section as S
where semester = 'Fall' and year = 2017 and
      exists (select *
              from section as T
              where semester = 'Spring' and year= 2018
              and S.course_id = T.course_id);
```

- **Correlation name** – variable S in the outer query
- **Correlated subquery** – the inner query



Use of “not exists” Clause

- Find all students who have taken all courses offered in the Biology department.

```
select distinct S.ID, S.name
from student as S
where not exists ( (select course_id
                    from course
                    where dept_name = 'Biology')
except
(select T.course_id
 from takes as T
 where S.ID = T.ID));
```

- First nested query lists all courses offered in Biology
 - Second nested query lists all courses a particular student took
- Note that $X - Y = \emptyset \Leftrightarrow X \subseteq Y$
- Note: Cannot write this query using = all and its variants



Test for Absence of Duplicate Tuples

- The **unique** construct tests whether a subquery has any duplicate tuples in its result.
- The **unique** construct evaluates to “true” if a given subquery contains no duplicates .
- Find all courses that were offered at most once in 2017

```
select T.course_id
from course as T
where unique ( select R.course_id
                  from section as R
                  where T.course_id= R.course_id
                      and R.year = 2017);
```



Subqueries in the From Clause



Subqueries in the From Clause

- SQL allows a subquery expression to be used in the **from** clause
- Find the average instructors' salaries of those departments where the average salary is greater than \$42,000."

```
select dept_name, avg_salary
from ( select dept_name, avg (salary) as avg_salary
      from instructor
      group by dept_name)
where avg_salary > 42000;
```

- Note that we do not need to use the **having** clause
- Another way to write above query

```
select dept_name, avg_salary
from ( select dept_name, avg (salary)
      from instructor
      group by dept_name)
      as dept_avg (dept_name, avg_salary)
where avg_salary > 42000;
```



With Clause

- The **with** clause provides a way of defining a temporary relation whose definition is available only to the query in which the **with** clause occurs.
- Find all departments with the maximum budget

```
with max_budget (value) as  
    (select max(budget)  
     from department)  
select department.name  
from department, max_budget  
where department.budget = max_budget.value;
```



Complex Queries using With Clause

- Find all departments where the total salary is greater than the average of the total salary at all departments

```
with dept_total (dept_name, value) as  
    (select dept_name, sum(salary)  
     from instructor  
     group by dept_name),  
dept_total_avg(value) as  
    (select avg(value)  
     from dept_total)  
select dept_name  
from dept_total, dept_total_avg  
where dept_total.value > dept_total_avg.value;
```



Scalar Subquery

- Scalar subquery is one which is used where a single value is expected
- List all departments along with the number of instructors in each department

```
select dept_name,  
        ( select count(*)  
          from instructor  
          where department.dept_name = instructor.dept_name)  
        as num_instructors  
from department;
```

- Runtime error if subquery returns more than one result tuple



Modification of the Database

- Deletion of tuples from a given relation.
- Insertion of new tuples into a given relation
- Updating of values in some tuples in a given relation



Deletion

- Delete all instructors

delete from *instructor*

- Delete all instructors from the Finance department

delete from *instructor*
where *dept_name* = 'Finance';

- *Delete all tuples in the instructor relation for those instructors associated with a department located in the Watson building.*

delete from *instructor*
where *dept name* in (**select** *dept name*
 from *department*
 where *building* = 'Watson');



Deletion (Cont.)

- Delete all instructors whose salary is less than the average salary of instructors

```
delete from instructor  
where salary < (select avg (salary)  
                from instructor);
```

- Problem: as we delete tuples from *instructor*, the average salary changes
- Solution used in SQL:
 1. First, compute **avg** (*salary*) and find all tuples to delete
 2. Next, delete all tuples found above (without recomputing **avg** or retesting the tuples)



Insertion

- Add a new tuple to *course*

insert into *course*

values ('CS-437', 'Database Systems', 'Comp. Sci.', 4);

- or equivalently

insert into *course* (*course_id*, *title*, *dept_name*, *credits*)

values ('CS-437', 'Database Systems', 'Comp. Sci.', 4);

- Add a new tuple to *student* with *tot_creds* set to null

insert into *student*

values ('3003', 'Green', 'Finance', *null*);



Insertion (Cont.)

- Make each student in the Music department who has earned more than 144 credit hours an instructor in the Music department with a salary of \$18,000.

insert into *instructor*

select *ID, name, dept_name, 18000*

from *student*

where *dept_name = 'Music' and total_cred > 144;*

- The **select from where** statement is evaluated fully before any of its results are inserted into the relation.

Otherwise queries like

insert into *table1 select * from table1*

would cause problem



Updates

- Give a 5% salary raise to all instructors

update *instructor*

set *salary* = *salary* * 1.05

- Give a 5% salary raise to those instructors who earn less than 70000

update *instructor*

set *salary* = *salary* * 1.05

where *salary* < 70000;

- Give a 5% salary raise to instructors whose salary is less than average

update *instructor*

set *salary* = *salary* * 1.05

where *salary* < (**select** **avg** (*salary*)
from *instructor*);



Updates (Cont.)

- Increase salaries of instructors whose salary is over \$100,000 by 3%, and all others by a 5%
 - Write two **update** statements:

```
update instructor  
  set salary = salary * 1.03  
  where salary > 100000;  
update instructor  
  set salary = salary * 1.05  
  where salary <= 100000;
```
 - The order is important
 - Can be done better using the **case** statement (next slide)



Case Statement for Conditional Updates

- Same query as before but with case statement

```
update instructor  
  set salary = case  
    when salary <= 100000 then salary * 1.05  
    else salary * 1.03  
  end
```



Updates with Scalar Subqueries

- Recompute and update *tot_creds* value for all students

```
update student S  
set tot_cred = (select sum(credits)  
                from takes, course  
                where takes.course_id = course.course_id and  
                    S.ID = takes.ID and  
                    takes.grade <> 'F' and  
                    takes.grade is not null);
```

- Sets *tot_creds* to null for students who have not taken any course
- Instead of **sum**(*credits*), use:

```
case  
  when sum(credits) is not null then sum(credits)  
  else 0  
end
```



End of Chapter 3